Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2001-II-Q08] | Part B (Q6-Q10)

Answer: 429

1. Since $2001 \in [3^6, 3^7]$, scaling gives $f(2001) = 729 - |2001 - 1458| = 186$.

2. On $[3^k, 3^{k+1}]$, the graph is a tent of height 3^k, specifically $f(x) = 3^k - |x - 2 \cdot 3^k|$. The first tent tall enough has $k = 5$ and height 243.

3. Solve $243 - |x - 486| = 186$: the first hit is $x = 429$, earlier than the other solution 543.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2018-II-Q10] | Part B (Q6-Q10)

Answer: 756

1. Every second iterate must be a fixed point. Partition the elements into fixed points, nonfixed elements mapping directly to fixed points, and the remaining elements, which must map to that middle level.

2. With $i \geq 1$ fixed points and j middle elements, choose levels in $\binom{5}{i}\binom{5-i}{j}$ ways; their images have $i^j j^{5-i-j}$ possibilities. This categorization is intrinsic, preventing double counting.

3. For $i = 1, 2, 3, 4, 5$, the totals are respectively $205, 380, 150, 20, 1$ (the last is the identity). They sum to 756.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2019-II-Q10] | Part B (Q6-Q10)

Answer: 547

1. Reduce 8θ modulo 180°. It is again acute and has the same entire sign itinerary; the stated uniqueness forces it equal θ. Thus 7θ is a multiple of 180°, leaving $\theta = 180k/7$ for $k = 1, 2, 3$.

2. For $k = 1$, the double angle still has positive tangent; for $k = 2$, the quadruple angle does. Both fail. For $k = 3$, the three residues are $540/7, 1080/7, 900/7$ degrees, respectively positive,negative,negative tangent, and repeat after multiplication by eight.

3. Therefore $\theta = 540/7$ and the answer is 547.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2020-II-Q08] | Part B (Q6-Q10)

Answer: 101

1. If $z \geq 0$ is a zero of f_{n-1}, its preimages are $n \pm z$; negative zeros give no preimages. With $T_j = j(j+1)/2$, induction shows that the zero set is the step-2 progression from $n - T_{n-1}$ to T_n.

2. The endpoints sum to $2n$, and the number of terms is $T_{n-1} + 1$. Hence the sum is $S_n = n(n(n-1)/2 + 1)$, an increasing expression.

3. $S_{100} = 495100$, while $S_{101} = 510151$. Therefore the first index is 101.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2000-I-Q12] | Part C (Q11-Q15)

Answer: 177

1. Composing reflections gives periods 1760 and 1056, hence also their gcd 352. The remaining reflection is $x \mapsto 46 - x$ modulo 352.

2. Among integer residues, this involution fixes 23 and 199 and pairs the other 350 residues. Thus there are $2 + 350/2 = 177$ possible classes; the given list covers every residue.

3. The function $f(x) = \cos(2\pi(x - 23)/352)$ satisfies all reflections and separates exactly these integer classes, proving the bound attainable.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2007-II-Q14] | Part C (Q11-Q15)

Answer: 676

1. The polynomial is nonconstant, and comparing leading coefficients makes it monic. If λ is a complex root, so is $2\lambda^3+\lambda$. A root of modulus greater than one would iterate to arbitrarily larger moduli, impossible for a finite root set.

2. The root product has modulus one because $f(0)=1$ and the polynomial is monic. Since every modulus is at most one, all must equal one. The image root must also have modulus one, so $|2\lambda^2+1|=1$ with $|\lambda|=1$, forcing $\lambda^2=-1$.

3. Real coefficients now give $f(x)=(x^2+1)^j$. Its functional identity is indeed satisfied. The value condition $5^j+10^j=125$ uniquely gives $j=2$. Therefore $f(5)=26^2=676$.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2009-I-Q13] | Part C (Q11-Q15)

Answer: 090

1. Subtract successive cleared recurrences to get $a_{n+2} - a_n = (a_{n+2} + 1)(a_{n+3} - a_{n+1})$. If one difference were nonzero, subsequent nonzero integer absolute differences would decrease by a factor of at least two forever, impossible.

2. Therefore the sequence has period two and $a_1 a_2 = 2009 = 7^2 \cdot 41$. Conversely any positive factor pair gives such a sequence.

3. The closest factors are $41, 49$, minimizing their sum at 90.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2009-II-Q14] | Part C (Q11-Q15)

Answer: 983

1. Normalize $u_n = a_n/2^n$. Then $u_{n+1} = (4/5)u_n + (3/5)\sqrt{1-u_n^2}$. Put $\sin\theta = 3/5, \cos\theta = 4/5$, with $30^\circ < \theta < 45^\circ$.

2. Starting at zero, normalized values are $\sin\theta, \sin 2\theta, \sin 3\theta$. Since $\cos 3\theta < 0$, the positive square root reverses the angle change, so subsequent values alternate $\sin 2\theta, \sin 3\theta$.

3. Thus $a_{10} = 2^{10}\sin 2\theta = 1024 \cdot 24/25 = 983.04$, whose floor is 983.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2012-II-Q11] | Part C (Q11-Q15)

Answer: 008

1. Write $f(x) = (6x - 7)/(9x + 3)$. Direct composition gives $f^{\circ 2}(x) = (-3x - 7)/(9x - 6)$ and $f^{\circ 3}(x) = x$ wherever the iterates exist.

2. As $1001 \equiv 2 \pmod 3$, the equation becomes $(-3x - 7)/(9x - 6) = x - 3$. Clearing the nonzero denominator yields $(3x - 5)^2 = 0$.

3. The candidate $5/3$ cycles through $5/3, 1/6, -4/3$, none of which is the pole $-1/3$, so all 1001 iterates exist. The answer is $5 + 3 = 8$.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2013-II-Q11] | Part C (Q11-Q15)

Answer: 399

1. Let the constant be a, and $S = f^{-1}(a)$. The nonempty set S contains a, since for any $t \in S$, $f(f(t)) = f(a) = a$. All members of S map to a, while every point outside S must map into $S \setminus \{a\}$.

2. For $|S| = k$, choose a in seven ways, the other $k-1$ members in $\binom{6}{k-1}$ ways, and outside images in $(k-1)^{7-k}$ ways. This construction is necessary and sufficient and determines S uniquely.

3. Thus $N = 7\sum_{k=1}^{7}\binom{6}{k-1}(k-1)^{7-k} = 7(0+6+240+540+240+30+1) = 7399$. The remainder is 399.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2021-II-Q15] | Part C (Q11-Q15)

Answer: 258

1. Put $k = \lceil \sqrt{n} \rceil$. Iterating to the next square gives $f(n) = k^2 + k - n$. The g iteration preserves parity, so if $n \equiv k \pmod 2$, it reaches the same square and $f(n)/g(n) = 1$. We therefore need opposite parity.

2. Then the first square of the parity of n is $(k+1)^2$, so $g(n) = f(n) + 2k + 2$. The ratio $4/7$ gives $f(n) = 8(k+1)/3$ and $n = k^2 + k - 8(k+1)/3$. Integrality requires $k \equiv 2 \pmod 3$. This formula always makes n even, so the opposite-parity condition requires odd k, hence $k \equiv 5 \pmod 6$.

3. The defining interval $(k-1)^2 < n \leq k^2$ has lower inequality equivalent to $k > 11$; its upper inequality is automatic. The least eligible integer k is therefore 17, yielding $n = 258$. This minimizes n because the square-root intervals are ordered by k.

4. Finally $17^2 - 258 + 17 = 48$ and $18^2 - 258 + 18 = 84$, giving $48/84 = 4/7$. Thus the least positive input is 258.

Source / 来源

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2025-II-Q13] | Part C (Q11-Q15)

Answer: 248

1. The substitution $y = (2x-1)/(x+1)$ conjugates the recurrence to $y_{j+1} = y_j(y_j - 1)$, with $y_1 = 13/12$. Thus $y_j = c/d$, where $d = 12^{2^{j-1}}$ and c stays coprime to 6.

2. Invert to get $x_j = (d+c)/(2d-c)$. Its numerator and denominator are positive and coprime: a common divisor divides both $3d, 3c$, hence divides 3, but neither term is divisible by 3. Therefore $m+n = 3d$.

3. For $j = 2025$, exponent $2^{2024} \equiv 16 \pmod{100}$. Modulo 125, $d \equiv 12^{16} = 41$; modulo 8, $d \equiv 0$. Chinese remainders give $d \equiv 416 \pmod{1000}$, so $m+n \equiv 248$.

Source / 来源